

Practice Final Exam

University of California, Santa Barbara

Time Allowed: 120 minutes — Topics: Purcell Chapters 1–4

Instructions:

- Show all work clearly. Box your final answers.
 - State all symmetry arguments before applying Gauss's law or integrating.
 - Scratch work without clear physical reasoning earns no credit.
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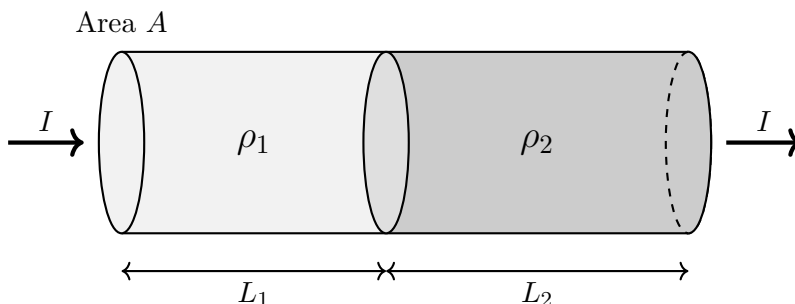
Problem 1: Gauss's Law (15 points)

A solid, uncharged, irregularly shaped chunk of metal is placed near a point charge $+q$. A student wants to find the electric field inside the metal. They draw a perfectly spherical Gaussian surface that encloses the point charge and exactly half of the metal chunk.

- (a) Is Gauss's Law ($\oint \vec{E} \cdot d\vec{a} = Q_{\text{enc}}/\epsilon_0$) mathematically **true** for this specific spherical surface? Briefly explain why or why not.
- (b) Can the student use this Gaussian surface to analytically compute the electric field at a point on the surface? Briefly explain why or why not.
- (c) Without using the integral, what is the exact electric field $\vec{E}$ at a point deeply embedded inside the metal chunk? Provide the fundamental physical reasoning (1-2 sentences).

Problem 2: The Two-Part Cylindrical Resistor (20 points)

A cylindrical resistor has a steady current I flowing uniformly through it from left to right. The material is uniform in cross-section (area A), but the left portion (length L_1 , uniform resistivity ρ_1) is made of a different material than the right portion (length L_2 , and higher resistivity ρ_2). Note: $\rho_1 < \rho_2$.



- What is the total power dissipated in this resistor in terms of the given quantities?
- Where, if anywhere, inside the conducting region of this resistor, is $\nabla \cdot \vec{J} \neq 0$? If you think it is non-zero somewhere, what is its sign (positive or negative)? Justify your answer.
- Where, if anywhere, inside the conducting region of this resistor, is $\nabla \cdot \vec{E} \neq 0$? If you think it is non-zero somewhere, what is its sign (positive or negative)? Justify your answer physically using boundary conditions.

Problem 3: Electrostatic Shielding & Uniqueness (25 points)

A thick, neutral conducting spherical shell has an inner radius a and an outer radius b . Inside the hollow cavity, a point charge $+q$ is placed. However, it is **not** placed at the center; it is shifted slightly to the right. A distance R away from the center of the shell ($R \gg b$), a second point charge $+Q$ is placed.

- Does the interior charge $+q$ experience a net electric force? If so, in what direction? Explain your reasoning.
- Is the surface charge density σ_a on the inner wall ($r = a$) uniformly distributed? What is the total charge integrated over this inner wall?
- Is the surface charge density σ_b on the outer wall ($r = b$) uniformly distributed? Explain why or why not using the Uniqueness Theorem.
- You now connect a grounding wire from the outer surface of the shell ($r = b$) to the earth, forcing $V = 0$ on the conductor. Does the force on the interior charge $+q$ change? Why or why not?

Problem 4: Capacitor Energy & Virtual Work (20 points)

An isolated parallel-plate capacitor has area A , initial separation s , and carries a constant charge $+Q$ and $-Q$ on its plates. Edge effects are negligible. You physically grab the plates and pull them apart until the new separation is $2s$.

- (a) Does the total energy stored in the electric field increase, decrease, or stay the same? Explain qualitatively.
- (b) Calculate the work you did with your hands to pull the plates apart, and show that it exactly equals the change in the stored field energy $U = \frac{\epsilon_0}{2} \int E^2 d\tau$.
- (c) Suppose an ideal electroscope was connected to the positive plate during this process. Would the leaves of the electroscope fall closer together, push further apart, or remain unchanged? Explain using the relationship between Q , C , and V .

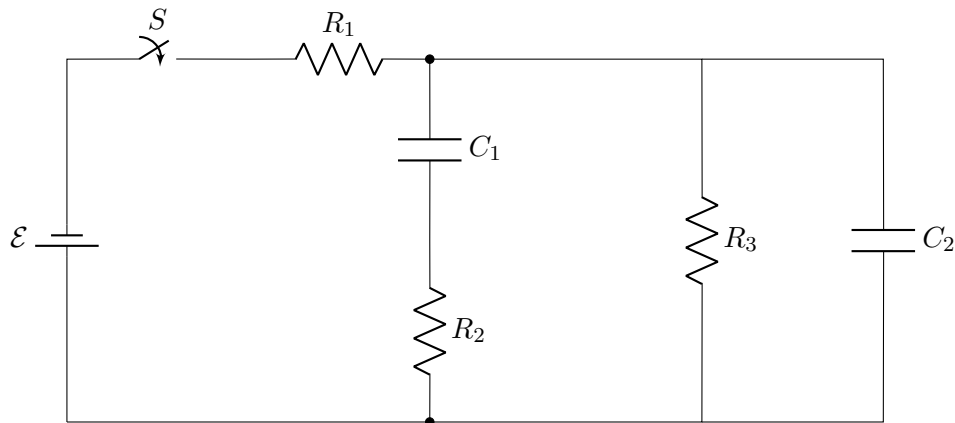
Problem 5: Method of Images (20 points)

An electric dipole consists of a charge $+q$ at $(0, d, d)$ and a charge $-q$ at $(0, -d, d)$. The entire xy -plane (at $z = 0$) is an infinite, grounded conducting sheet ($V = 0$).

- (a) Describe the configuration of fictitious image charges required to solve for the potential in the region $z > 0$. Provide their charges and (x, y, z) coordinates.
- (b) What is the total work required to assemble the *original* real dipole (bringing $+q$ and $-q$ from infinity to their final positions) in the presence of this grounded plane? Set up the expression (you do not need to fully simplify the algebraic fractions, but all terms and distances must be clearly defined).

Problem 6: The Asymptotic RC Trap (20 points)

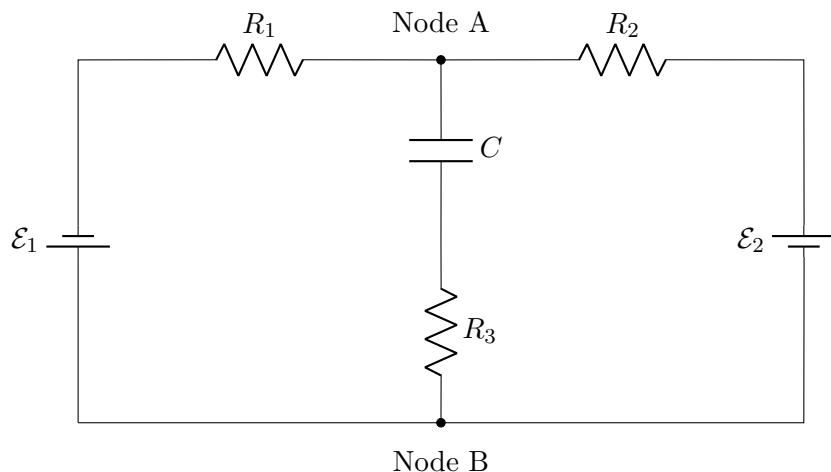
Consider the circuit shown below. The switch S has been open for a long time, and all capacitors are initially uncharged. At time $t = 0$, the switch is closed.



- Immediately** after the switch is closed (at $t = 0^+$), what is the total current I flowing out of the battery? Explain your reasoning physically.
- After a very long time ($t \rightarrow \infty$), what is the total current I flowing out of the battery?
- In the steady state ($t \rightarrow \infty$), what is the exact potential difference across capacitor C_1 , and what is the potential difference across capacitor C_2 ?
- A student attempts to calculate the total power dissipated by the circuit at $t \rightarrow \infty$ and writes $P = \mathcal{E}^2 / (R_1 + R_2 + R_3)$. Is this equation mathematically **true**, **false**, or **useful but incomplete**? Briefly explain the conceptual error.

Problem 7: The Steady-State Bridge (20 points)

In the circuit below, two ideal batteries ($\mathcal{E}_1$ and $\mathcal{E}_2$) oppose each other in a multi-loop network. A capacitor C sits in the middle branch. The circuit has been connected for a very long time and has reached a complete steady state.



- (a) A student starts writing a 3×3 matrix of Kirchhoff's Current and Voltage Laws to find the current through R_1 . Explain why this approach, while technically valid, ignores the physical reality of the steady state.
- (b) Calculate the steady-state current flowing through R_1 .
- (c) Determine the total charge Q stored on the capacitor C in the steady state. Which plate (top or bottom) holds the positive charge? Assume $\mathcal{E}_1 > \mathcal{E}_2$.
- (d) Suppose the capacitor C suffers a catastrophic dielectric breakdown and fuses into a solid block of conductive metal (acting as an ideal, resistanceless wire). Set up, **but do not solve**, the Kirchhoff loop and junction equations required to find the new current flowing through the middle branch.